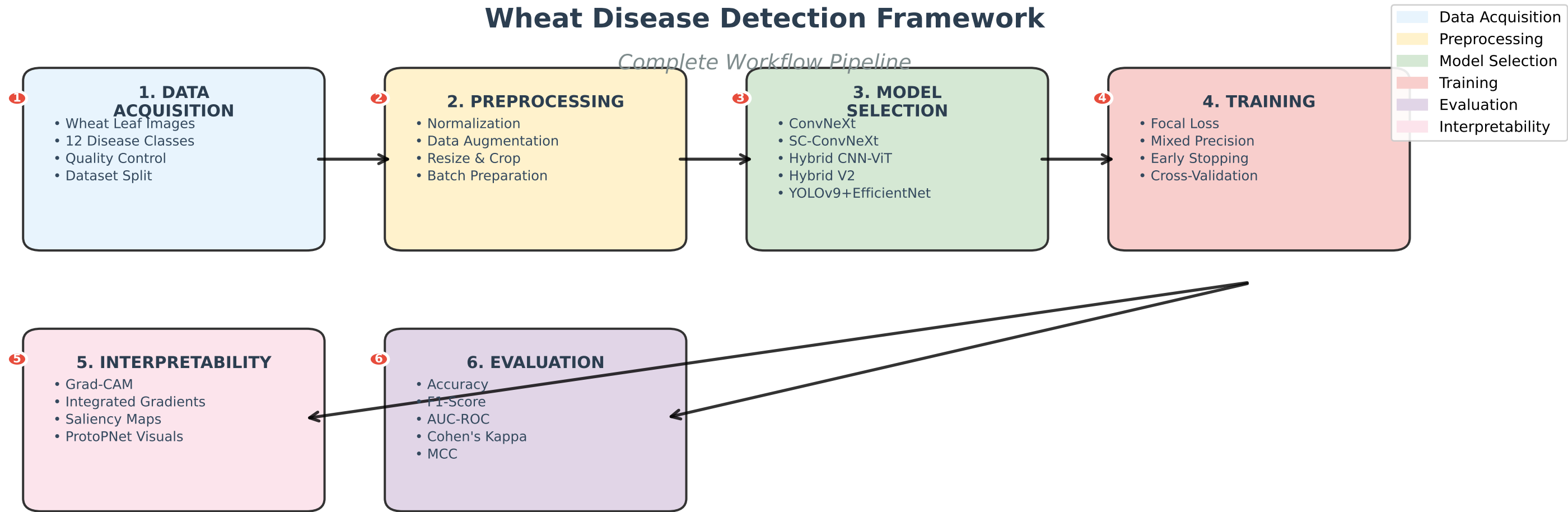


Wheat Disease Detection Framework



Performance Results

Best Model: ConvNeXt (90.41% accuracy)
F1-Score Range: 70.84% - 89.99%
Training Time: 2.1 - 5.5 hours
Model Size: 15.2M - 52.3M parameters

Technical Infrastructure

Hardware: NVIDIA Tesla T4 (16GB VRAM) | Intel Xeon E5-2686 v4
Software: PyTorch 1.12.1 | CUDA 11.8 | OpenCV 4.6.0
Dataset: 12 wheat disease classes | Stratified 70/15/15 split
Reproducibility: Fixed seeds | CUDA determinism | Bootstrap validation